

A Reduced BM Four-Frame from Time, Space, Energy and Momentum

A Spectral-Geometric Perspective on Prime Structures

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Abstract

We propose a reduced four-dimensional framework, referred to as the Bamberg Model (BM), consisting of four distinguished coordinates

$$(A, C, e, b)$$

interpreted heuristically as time, space, energy and momentum.

Within this setting a Minkowski-type quadratic form emerges naturally and allows the construction of a Dirac-type operator

$$D_{\text{BM}} = \Gamma_A A + \Gamma_C C + \Gamma_e e + \Gamma_b b.$$

The square of this operator reproduces a relativistic-like invariant combining space–time and energy–momentum structures.

Motivated by the Hilbert–Pólya idea we explore the possibility that arithmetic structures of primes may produce Dirac-like spectral behaviour related to the imaginary parts of the nontrivial Riemann zeros.

The work is presented as a heuristic spectral-geometric program linking logarithmic prime structures, Dirac operators and numerical experiments.

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1 Introduction

The relationship between prime numbers and the nontrivial zeros of the Riemann zeta function remains one of the central problems of mathematics.

The Hilbert–Pólya conjecture proposes that the imaginary parts of the Riemann zeros

$$\rho_n = \frac{1}{2} + i\gamma_n$$

arise as eigenvalues of a suitable self-adjoint operator.

At the same time developments in spectral geometry have demonstrated that geometric information may be encoded in the spectrum of Dirac operators.

The present article introduces a reduced framework, the Bamberg Model (BM), which combines arithmetic prime structures with a four-dimensional geometric setting.

2 The BM Four-Frame

We introduce four coordinates

$$(A, C, e, b)$$

interpreted heuristically as

symbol	interpretation
A	temporal coordinate
C	spatial coordinate
e	energy-like variable
b	momentum-like variable

The pair (A, C) forms a reduced space–time structure while (e, b) represents an energy–momentum analogue.

3 Reduced Minkowski Structure

Assuming A to be time-like and C space-like we define the quadratic form

$$ds^2 = c_M^2 dA^2 - dC^2.$$

This yields a light-cone structure

$$\left| \frac{dC}{dA} \right| \leq c_M.$$

4 Energy–Momentum Structure

The energy–momentum analogue of the interval is

$$e^2 - c_M^2 b^2.$$

This leads to the invariant

$$e^2 - c_M^2 b^2 = m_{\text{BM}}^2 c_M^4.$$

5 Clifford Structure

The four BM coordinates naturally generate a Clifford algebra

$$Cl(1, 3).$$

6 BM Dirac Operator

We define

$$D_{\text{BM}} = \Gamma_A A + \Gamma_C C + \Gamma_e e + \Gamma_b b.$$

6.1 Proposition

$$D_{\text{BM}}^2 = A^2 - C^2 + e^2 - b^2.$$

This expression combines space–time and energy–momentum structures.

7 Prime Structures

The multiplicative group

$$(\mathbb{Z}/12\mathbb{Z})^\times = \{1, 5, 7, 11\}$$

is isomorphic to the Klein four group

$$V_4.$$

These residue classes define four arithmetically distinguished directions.

8 Logarithmic Prime Coordinates

Multiplicative structures become additive under

$$x_p = \log p.$$

This allows prime data to be embedded into an additive geometric space.

9 A Dirac Operator on the Prime Graph

Let V denote the set of primes.

Define a graph connecting primes belonging to the residue classes

$$\{1, 5, 7, 11\} \pmod{12}.$$

The operator

$$(Df)(p) = \sum_{q \sim p} (\log p - \log q) f(q)$$

defines a discrete Dirac-type operator.

10 Numerical Experiments

We consider prime quadruples

$$(p_1, p_2, p_3, p_4)$$

with residues

$$\{1, 5, 7, 11\} \pmod{12}.$$

Coordinates are defined as

$$C = \log(p_1 p_2 p_3 p_4)$$

and

$$b = \log(p_1^2 + p_2^2 + p_3^2 + p_4^2).$$

The BM spectral value is

$$\lambda = \sqrt{|C^2 - b^2|}.$$

These values are compared numerically with Riemann zero ordinates γ_n .

11 Relation to the Explicit Formula

The Riemann explicit formula relates prime numbers and zeros via

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \log(2\pi) - \frac{1}{2} \log(1 - x^{-2}).$$

The appearance of logarithmic prime contributions

$$\log p$$

matches naturally the BM coordinate $x_p = \log p$.

12 Possible Relation to an E_8 Geometry

Introducing dual coordinates

$$(A^*, C^*, e^*, b^*)$$

extends the BM space to

$$\mathbb{R}^8.$$

Such eight-dimensional structures are reminiscent of highly symmetric lattices such as the E_8 lattice.

While no direct identification is claimed, the dimensional extension suggests that the BM framework may admit an embedding into higher-dimensional spectral geometry.

13 Discussion

The BM framework should be understood as a heuristic research program.

The numerical experiments do not establish a spectral identity with the Riemann zeros but suggest that arithmetic prime structures may generate Dirac-like spectral behaviour.

14 Outlook

Future work may include

- rigorous construction of a BM Dirac operator
- deeper statistical comparison with Riemann zeros
- connections with spectral geometry and noncommutative geometry

References

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